

MADISON EMSHOUSEN

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PROFESSIONAL SUMMARY

Results-driven Data Analyst and ML Engineer with 2+ years of experience building machine learning pipelines, automating investigative workflows, and delivering data-driven insights at enterprise scale. Proven ability to leverage Python, SQL, and LLMs to protect high-value assets, support complex investigations, and inform strategic decision-making. Adept at cross-functional collaboration, stakeholder communication, and translating raw data into actionable intelligence.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, C#, R, Lua, HTML
ML / AI: Machine Learning, Large Language Models (LLMs), Predictive Modeling, Anomaly Detection, NLP, MLOps
Data & Analytics: ETL Pipelines, Data Engineering, Dashboard Development, Data Visualization, Matplotlib, Pandas
Databases: SQL (advanced), RDBMS, Query Optimization, Indexing, React.js
Security & Investigations: Digital Forensics, Fraud Detection, Incident Response, OSINT, Chain of Custody, Threat Analysis
Tools & Platforms: Jupyter, Git/GitHub, Tableau/BI tools, Zoom, Agile

EXPERIENCE

Control Risks @ Meta Oct 2024 – Present
Data Analyst — Investigation Enablement, Global Security & Investigations Remote

- Engineered end-to-end ML pipelines in Python and SQL to track and recover over **\$200M** in lost and stolen Meta devices, enabling real-time asset recovery and reducing financial exposure at enterprise scale.
- Conducted forensic SQL investigations into organized retail theft incidents, identifying repeat offenders across multiple break-in events by cross-referencing behavioral patterns and transaction records in complex Meta databases.
- Designed and deployed LLM-powered dashboards to surface trends and generate actionable intelligence, reducing manual analyst review time and improving investigative throughput.
- Automated data collection and end-to-end investigative workflows using custom Python scripts, eliminating repetitive manual processes and accelerating case resolution time.
- Collaborated with cross-functional teams, including legal, security operations, and engineering, to implement scalable, data-driven solutions supporting global investigations.
- Presented findings and intelligence reports to senior stakeholders, driving informed decision-making on asset protection strategy.

Codakid Oct 2022 – Oct 2024
Computer Science Instructor Remote

- Delivered personalized 1:1 computer science instruction via Zoom to 50+ students across skill levels, specializing in game development using Python, C#, Lua, Java, HTML, and JavaScript.
- Designed adaptive curriculum and lesson plans tailored to individual student learning styles and proficiency levels, achieving consistent improvement in student performance metrics.
- Tracked student progress and delivered detailed post-session performance reports, enabling data-driven optimization of learning outcomes and identifying areas for improvement.

PROJECTS

Online Music Database | *SQL, Java, React.js*

- Designed and implemented a scalable RDBMS with optimized backend architecture supporting music playback, playlist management, and user interaction; applied advanced indexing and query optimization techniques to maximize performance under load.

Project Helios | *Python, Matplotlib, Data Visualization*

- Led a data visualization and analysis project using Python and Matplotlib to process solar flare datasets, uncover temporal trends, and deliver interactive visual reports with actionable insights.

EDUCATION

University of Houston Class of 2024
Bachelor of Science in Computer Science — Houston, Texas
Relevant Coursework: Data Structures & Algorithms (C++), Digital Image Processing (Python), Data Science I & II (Python, R), Cybersecurity (Python), Databases (C#, React.js, SQL)

ACHIEVEMENTS

CodeRed UH Hackathon Winner — University of Houston 2025
CodeCoogs Team Lead — Led a student software engineering organization 2024
CodeCoogs Member — Active contributor to CS community 2022 – 2024

VOLUNTEER EXPERIENCE

Computer Science Association — Houston City College Aug 2025 – Present
Mentor students in CS fundamentals; help organize workshops and events connecting peers with industry professionals.